

Coding & Solution

Date	08 October 2026
Team ID	SWTID-2026-2386
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Dhanam0706/EduGenie
Programming Language(s)	Python 3.10+, JavaScript, HTML, CSS
Framework(s) Used	FastAPI, Jinja2, Pydantic; Google Gemini via <code>google-generativeai</code>
Key Features Implemented	Question answering with follow-ups; structured concept explanation; summarization; quiz generation (3 questions, 4 options, JSON cleaned and validated); beginner-to-advanced learning path; automatic Gemini model fallback; input validation; friendly error handling; API key kept in <code>.env</code> ; Dockerfile; automated tests that fake Gemini; load test
Pending / Incomplete Features	Voice input, multilingual support, mobile app, progress tracking, PDF/image input (see Scalability & Future Plan)
Setup / Run Instructions	<code>pip install -r requirements.txt</code> , copy <code>.env.example</code> to <code>.env</code> and add <code>GEMINI_API_KEY</code> , then <code>uvicorn main:app --reload</code> and open http://localhost:8000 . Full steps in README.md

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Team to confirm after the first full run with a real key (see README)
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

- Earlier descriptions of this project name Gemini 1.5 Pro. That model family is retired, so the model is configured with `GEMINI_MODEL` (default `gemini-3.8-flash`) and the app falls back through other current models automatically. Check <https://ai.google.dev/gemini-api/docs/models> if a model is retired.
- The original plan used a local LaMini-Flan-T5 model for explanations. Explanations now use Gemini too, so the project needs no large local model, torch or transformers.

- Generated text is always shown with `textContent`, and the quiz JSON is validated on the server before it is shown.